

Exercise-2 Factory Method Pattern:

Document.java:

```
public interface Document {  
    void open();  
}
```

WordDocument.java:

```
public class WordDocument implements Document {  
  
    @Override  
    public void open() {  
        System.out.println("Opening Word Document...");  
    }  
}
```

PdfDocument.java:

```
public class PdfDocument implements Document {  
  
    @Override  
    public void open() {  
        System.out.println("Opening PDF Document...");  
    }  
}
```

ExcelDocument.java:

```
public class ExcelDocument implements Document {  
  
    @Override  
    public void open() {  
        System.out.println("Opening Excel Document...");  
    }  
}
```

DocumentFactory.java:

```
public abstract class DocumentFactory {  
  
    public abstract Document createDocument();  
  
    public void openDocument() {  
        Document document = createDocument();  
        document.open();  
    }  
}
```

WordFactory.java:

```
public class WordFactory extends DocumentFactory {  
  
    @Override  
    public Document createDocument() {
```

```
        return new WordDocument();
    }
}
```

PdfFactory.java:

```
public class PdfFactory extends DocumentFactory {

    @Override

    public Document createDocument() {

        return new PdfDocument();

    }
}
```

ExcelFactory.java:

```
public class ExcelFactory extends DocumentFactory {

    @Override

    public Document createDocument() {

        return new ExcelDocument();

    }
}
```

FactoryMethodDemo.java:

```
public class FactoryMethodDemo {

    public static void main(String[] args) {

        DocumentFactory wordFactory = new WordFactory();

        wordFactory.openDocument();

    }
}
```

```
DocumentFactory pdfFactory = new PdfFactory();
```

```
pdfFactory.openDocument();
```

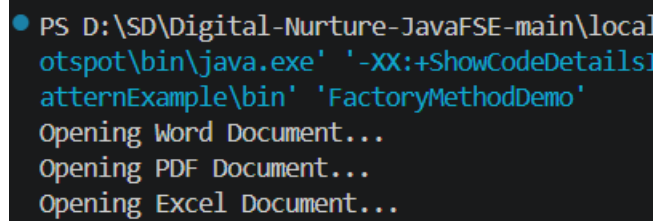
```
DocumentFactory excelFactory = new ExcelFactory();
```

```
excelFactory.openDocument();
```

```
}
```

```
}
```

Output:

A terminal window with a dark background and light blue text. It shows a command to run a Java application with specific flags, followed by three lines of output indicating document opening status.

```
PS D:\SD\Digital-Nurture-JavaFSE-main\local  
otspot\bin\java.exe' '-XX:+ShowCodeDetails:  
atternExample\bin' 'FactoryMethodDemo'  
Opening Word Document...  
Opening PDF Document...  
Opening Excel Document...
```